

typst:

```
$
&not (p and (q or r)) \
&= not p or (not q and not r) & "(demorgan twice)" \
&= (not p or not q) and (not p or not r) wide &
"(distributive)" \
&= not (p and q) and not(p and r) & "(demorgan)"
$
```

latex:

```
\begin{aligned}
& \neg (p \wedge (q \vee r)) \\
&= \neg p \vee (\neg q \wedge \neg r) \quad \text{(demorgan twice)} \\
&= (\neg p \vee \neg q) \wedge (\neg p \vee \neg r) \quad \text{(distributive)} \\
&= \neg(p \wedge q) \wedge \neg(p \wedge r) \quad \text{(demorgan)}
\end{aligned}
```

$$\begin{aligned}
& \neg(p \wedge (q \vee r)) \\
&= \neg p \vee (\neg q \wedge \neg r) && \text{(demorgan twice)} \\
&= (\neg p \vee \neg q) \wedge (\neg p \vee \neg r) && \text{(distributive)} \\
&= \neg(p \wedge q) \wedge \neg(p \wedge r) && \text{(demorgan)}
\end{aligned}$$